

機関番号	研究種目番号	応募区分番号	小区分	整理番号
82512	06	1	07040	0002

令和9(2027)年度 基盤研究（C）（一般）研究計画調書

令和 8 年 9 月 8 日
版

新規

研究種目	基盤研究(C)	応募区分	一般
小区分	経済政策関連		
研究代表者 氏名	(フリガナ) イトウ セイロウ (漢字等) 伊藤 成朗		
所属研究機関	独立行政法人日本貿易振興機構アジア経済研究所		
部 局	開発研究センター		
職	主任調査研究員		
研究課題名	Production approach to minimum wage impacts		
研究経費 〔千円未満の 端数は切り 捨てる〕	年度	研究経費 (千円)	使用内訳 (千円)
			設備備品費 消耗品費 旅費 人件費・謝金 その他
	令和9年度	1,650	400 0 1,100 0 150
	令和10年度	150	0 0 0 0 150
	令和11年度	2,050	400 0 1,000 0 650
	令和12年度	1,150	0 0 1,000 0 150
	令和13年度	0	0 0 0 0 0
	総計	5,000	800 0 3,100 0 1,100
開示希望の有無	審査結果の開示を希望する		
研究計画最終年度前年度応募	--		

研究組織（研究代表者及び研究分担者）

	氏名（年齢）	所属研究機関 部局 職	学位 役割分担	令和9年度 研究経費 （千円）	エフ ォ ー ト （%）
研究代表者	50450482（58） イトウ セイロウ 伊藤 成朗	独立行政法人日本貿易振興機構アジア経済研究所 開発研究センター 主任調査研究員	Ph.D. (Economics) 研究代表者	1,650	30
合計 1 名			研究経費合計	1,650	

1. Research Objectives, Research Method, etc.

This research proposal will be reviewed in the Basic Section of the applicant's choice. In filling this application form, refer to the Application Procedures for Grants-in-Aid for Scientific Research -KAKENHI-.

Research objectives, research method, etc. should be described within 4 pages.

A succinct summary of the research proposal should be given at the beginning.

The main text should give descriptions, in concrete and clear terms, of (1) scientific background of the proposed research, the circumstances leading to the conception of the proposed research, and the "key scientific question" comprising the core of the research plan, (2) the purpose, scientific originality, and creativity of the research project, (3) research trends in the relevant area and the positioning of this research in the area, (4) what will be elucidated, and to what extent and how will it be pursued during the research period, (5) preparation status towards achievement of the purpose of the research project, and (6) what sort of internationality this proposed research has (e.g. it will be a future world-leading research project, it will contribute to the development of global research projects through collaboration, it will create high value as a research project unique to Japan, etc.).

If the proposed research project involves Co-Investigator(s) (Co-I(s)), a concrete description of the role-sharing between the Principal Investigator (PI) and the Co-I(s) should be given.

[SUMMARY]

Minimum wages are a popular policy to reduce poverty and inequality. It is also a difficult policy because the low-wage workers may lose their jobs. We test the leading disemployment mechanism: Larger labour market power leads to smaller disemployment impacts of minimum wages. What distinguishes our research is the use of the production approach to estimate *wage markdowns*, (marginal value product of labour)/(wage), as a measure of labour market power. This addresses the limitations of Herfindahl-Hirshman index (HHI) of employment shares used in previous studies: HHI is sensitive to how one chooses the local labour market boundaries. The production approach requires only output elasticities and cost shares of labour and material inputs, which can be obtained from corporate tax returns data of South African Revenue Services (SARS). For South Africa (SA), we will show:

A. Distribution of labour market power across geography and industry over time

B. Whether oligopsony reduces the disemployment impacts of minimum wages

A. and the use of wage markdown in B. in a developing country context are limited. Findings from B. will inform more effective minimum wage policies and A. will form the basis of labour market power empirics.

[MAIN TEXT]

(1) Scientific background of the proposed research, the circumstances leading to the conception of the proposed research, and the "key scientific question" comprising the core of the research plan

Background: *Is it reasonable to raise the minimum wages for more than 50% when the unemployment rate is massively high at 25%?* Policymakers in South Africa had to decide on such a difficult question without credible guidance from economists. After the early literature debates, it is known that the *overall* disemployment impacts are relatively small, and economists are asking why (Manning, 2021). Economists traditionally invoked oligopsony models (Robinson, 1933, Bhaskar et al., 2002, Manning, 2011). According to the model, an increase in minimum wages has two opposing effects on employment: It reduces employment due to increased costs of employment, and it increases employment through lower recruitment costs due to greater retention of workers and an increased number of job searchers. Therefore, combined effects can result in small negative, null, or even small positive impacts (Manning, 2006). If the labour market power differs across firms, the overall impact masks larger disemployment impacts of firms with smaller power. *We can reduce the disemployment impacts by informing the regions of less concentrated labour markets to upgrade the market infrastructure to make job search and turnover costs smaller, or give temporary support to firms with smaller power to reorganize rather than shutting down.*

Conception of proposed research: While being the primary theory proposed as an explanation, there are only a few papers empirically testing the hypothesis (Azar et al., 2024; Popp, 2024; Faia et al., 2026), and almost none in the developing country context except the applicant's (ongoing research ending this year). These works provide new insights, but they also share a limitation except for Faia et al., 2026: HHI is considered confounded and its measurement is boundary sensitive. To address this, the proposed research draws on the production approach to estimate wage markdown, the direct *definition* of labour market power. It requires only tax returns data and removes the boundary choice, giving a direct measure rather than a confounded proxy.

Key scientific questions, Purpose, Scientific originality and creativity:

- Empirics of labour market power
 - **Purpose = To answer the key question:** How is labour market power distributed across geography and across industries over time?
- Labour market power on disemployment impacts of minimum wages
 - **Purpose = To answer the key question:** Does larger labour market power reduce disemployment impacts of minimum wages?

(2) Purpose, scientific originality, and creativity of the research project

Purpose: A. To obtain distributions of labour market power across geography and industries over time; B. To test if labour market power reduces the disemployment impacts of minimum wages.

Scientific originality and creativity: First firm-level wage-markdown estimation in a developing country + First direct (non-proxy) test of the oligopsony–minimum-wage mechanism in a developing country.

Explanation: Wage markdown is the definition of labour market power. The oft used HHI is useful but comes with two limitations. (a) It is an indirect measure: HHI is an outcome of labour market power and not the power itself. One must control confounding factors, such as population density and firm productivity, in its relationship with wages and employment. For example, less-productive rural areas have higher HHIs and less capacity to absorb a minimum-wage increase, inducing a positive correlation between HHI and disemployment impacts (Azar et al., 2024). Controlling these is data demanding: density is measured only every four–five years (ill-suited to

[1. Research Objectives, Research Method, etc. (continued from the previous page)]

high-mobility SA), and productivity needs a separate panel estimation. (b) HHI can be different by the choice of area units in defining local labour markets: Main Place (most granular), Local Municipality, District Municipality, Province (most aggregated). Therefore, one must decide which level of area unit is relevant, by implicitly taking a stand on the degree of local labour market integration which is chosen from the four administrative area units.

The production approach does not require separate production panel data and gives wage markdown at the firm level using corporate accounting panel data alone without a reference to local labour markets (see below).

(3) Research trends in the relevant area and the positioning of this research in the area

Trends: The proposed research has two roots. First, it builds on the production approach to output markups (Hall, 1988; De Loecker and Warzynski, 2012), which is extended to input price markdowns (Morlacco, 2019; Yeh et al., 2023, Syverson, 2025). Brooks et al. (2021) apply it to manufacturing firm data of China and India, Amodio et al. (2025) apply to an 82 harmonized developing country data set under limited sample size in each country, both following Akerberg et al. (2015). Recent studies that apply to firm survey data of China (Pham, 2023) and Mexico (Estefan et al., 2024) use the improvements of Gandhi et al. (2020) over Akerberg et al. (2015). Importantly, the method of Pham (2023) requires only revenue and input cost data, and a materials price deflator to estimate elasticities. Other extensions include input price markdowns under Leontief-in-materials production technology (Rubens, 2023) and under various modes of labour market competitions (Delavastita and Rubens, 2025). Amodio et al. (2026) use “the monopsonist approach” to wage markdown [see **Method** equation (1) below].

The second root is the minimum wage literature. While disemployment impacts have been debated empirically for decades, its mechanism assessment began only recently. Azar et al. (2024) is the first to show the relationship between disemployment impacts and labour market power proxied by HHI, which affirms the hypothesis on a population of stock clerks, retail salespersons, and cashiers in the US online labour market. The applicant’s ongoing research is an application of Azar et al. (2024) on a different population (developing country, agricultural producers). Faia et al. (2026), under the 2015 German minimum wage introduction, use the production approach and derive a theory-consistent estimation specification which we will also examine for comparability.

Positioning: We use the state-of-the-art method from the first (production approach) to the second (minimum wage research). This combination is novel in a developing country context and the findings will be informative in poverty/inequality reduction. Outputs from the first forms the basis of future labour market power empirics which can also be combined with other socioeconomic indicators (such as local Gini coefficients or unemployment rates computed from the *Labour Force Survey* to understand the outreach of labour market power effects).

Method: Morlacco (2019, equation 8) showed wage markdowns are given as a ratio of output elasticities and cost shares (ratio of θ_X/α^X and θ_M/α^M) under material M market competition.

Consider the monopsonist profit maximization on materials M and labour X with production function $Q = f(M, X)$, materials price function $D(M)$, wage function $W(X)$, and revenue function $R(Q)$:

$$\pi = R\{f(M, X)\} - D(M)M - W(X)X$$

The FOC of labour is

$$R'(Q)f_X(M, X) = W(X) + W'(X)X = W(X)\left(1 + W'(X)\frac{X}{W(X)}\right) = W(X)(1 + \varepsilon^{-1}).$$

This gives the monopsonist approach to markdown μ^X :

$$\mu^X \equiv \frac{MRPX}{W(X)} = \frac{R'(Q)f_X(M, X)}{W(X)} = 1 + \varepsilon^{-1} \quad (1)$$

(needing the labour supply elasticity---hard to measure and usually a dead end). Production approach to markdown is more empirically friendly:

$$\mu^X \equiv \frac{R'(Q)f_X(M, X)}{W(X)} = \frac{R'(Q)}{W(X)} \frac{\theta_X Q}{X} = \frac{P(1-\eta^{-1})}{W(X)} \frac{\theta_X Q}{X} = (1 - |\eta^{-1}|) \frac{\theta_X}{\alpha^X} \quad (2)$$

Production approach to markdown in (2) used output elasticity of X for the second equality, $\theta_X \equiv f_X(M, X) \frac{X}{Q} \Rightarrow f_X(M, X) = \frac{\theta_X Q}{X}$, and the usual FOC from monopolist profit maximization where η is output demand elasticity of price for the third equality:

$$R'(Q) = MR = P(1 - \eta^{-1}) = MC. \quad (3)$$

The fourth equality used (3), X ’s cost share in revenue $\alpha^X \equiv W(X)X/PQ$ and $\eta = |\eta|$. Using (3) again and define price markup $\mu \equiv P/MC$ inversion, or $\mu^{-1} \equiv \frac{MC}{P} = 1 - \eta^{-1} = 1 - |\eta^{-1}|$. Substituting to (2):

$$\frac{\theta_X}{\alpha^X} = \mu^X \mu. \quad (4)$$

We get a similar expression for M . If we assume the materials market is not oligopsonistic, i.e., $\mu^M = 1$, then,

$$\frac{\theta_M}{\alpha^M} = \mu^M \mu = \mu, \quad (5)$$

so taking a ratio of (4) over (5) gives μ^X . A nice aspect of this approach is forming a ratio cancels out the

[1. Research Objectives, Research Method, etc. (continued from the previous page)]

hypothetical monopolist markup [or more generally, output conduct $R'(Q) = z(Q)P$ and still get (4) - (5)], so the monopolist conduct is a convenience of exposition, and its plausibility does not matter for markdown estimation. We use the following event-study difference-in-differences (DID) estimators:

$$X_{it} = a_i + a_t + b * FA_i + c * \mu_i^X + d_t * FA_i \mu_i^X + e_{it}, \quad (6)$$

$$X_{it} = a_i + a_t + b * FA_i + c * \ln(\mu_i^X)^{-1} + d_t * FA_i \ln(\mu_i^X)^{-1} + e_{it}. \quad (7)$$

$a_i, a_t, FA_i, \mu_i^X, e_{it}$ are firm, period fixed effects, fraction affected (sub-minimum wage workers/total workers), wage markdown, and random errors for firm i in period t . The second specification (7) follows Faia et al. (2026).

Threats to identification

$\mu^M = 1$? The assumption $\mu^M = 1$ is necessary only if one is interested in the level of μ^M , which we are not. If μ^M is uniform within a sector, ratios across firms still recover relative markdowns. We can choose the firm with lowest μ^X , set its $\mu^X = 1$, and estimate c, d_t as the relative magnitude. Second, since $\mu^M \geq 1$, μ^X / μ^M is a lower bound on μ^X . The data we use allows us to test the assumption $\mu^M = 1$ directly: buyer zip codes in the procurement portion of the data let us recover the supplier concentration each firm faces.

$\text{cov}[\mu_i^X, e_{it}] = 0$? High-exposure (high FA_i) firms tend to be labour-intensive (high α_i^X) (or $\text{cov}[FA_i, \alpha_i^X] > 0$), so $\mu_i^X = \theta_{Xi} / \alpha_i^X$ can be low, while they may have lower employment (smaller e_{it} when X is employment) by a minimum wage increase. This can create a positive correlation between markdown μ_i^X and e_{it} , which can mechanically shrink p values. While there is no theoretically *a priori* reason that a large α_i^X translates one-to-one to low μ_i^X (or $\text{cov}[\mu_i^X, \alpha_i^X] < 0$), because θ_{Xi} may also be larger, we can address such concern by testing it. If we find within-sector covariances $\text{cov}[FA_i, \alpha_i^X] \approx 0$ or $\text{cov}[\mu_i^X, \alpha_i^X] \approx 0$, more confidence on $\text{cov}[\mu_i^X, e_{it}] \approx 0$.

$\theta_X / \alpha^X = \text{labour market power}$? θ_X / α^X is a labour wedge that captures any input distortion (Hashemi et al., 2022); we read it as market power, because concentration and turnover costs from distortions can isolate the firm from the outside market which leaves them to exercise power.

Data requirement and applications: This method only requires output elasticities of inputs θ_M, θ_X and cost shares α^M, α^X , both of which can be computed from firm accounting data at the firm level and a materials price deflator, and does not require supply elasticity of labour as in “the monopsonist approach.” We use administrative data which cover the universe of firms that file tax returns at South African Revenue Services (SARS). To estimate output elasticities of input, we use the methods of Raval (2023) and Pham (2023).

(4) What will be elucidated, and to what extent and how will it be pursued during the research period

What:

The proposed research will show, in South Africa:

- A. Distribution of labour market power across geography and industries over time
 - B. Results of a direct test if the labour market power mitigates the disemployment impacts of minimum wages
- A. is an input to B. However, most developing-country studies of labour market power use strong assumptions (Brooks et al., 2021; Amodio et al., 2025) or other methods (Amodio et al., forthcoming), so A. has merits of its own. B. identifies the mechanism within agriculture; its economy-wide generalisation is a natural next step, which the economy-wide markdown map in A. helps to frame. We pursue these outcomes in two separate research papers.

What extent:

A. Wage markdown: The data cover the universe of firms which are filing tax returns in SA for the period of 2009-2022. Wage markdowns can be estimated for this coverage.

B. Oligopsony effects on disemployment impacts of minimum wages: Disemployment impacts of minimum wage can only be examined before and after minimum wage changes. We focus on the sharp 52% increase in minimum wage in 2013 for the event study. Given the policy was universally introduced in agriculture, there is no within-sector control group. We follow Harasztosi and Lindner (2019) in defining the treatment variable as the degree of exposure (fraction of subminimum wage workers) to the minimum wage increase and perform event-study estimation on % change in the subsequent employment. The identification assumption is the common trend in all values of exposure as in the continuous DID estimator (Callaway et al., 2025), and the applicant’s ongoing research shows that it holds for pre-periods.

How:

SARS data sets are confidential information. The applicant already has the access privilege. The applicant will visit the data site in SA to process estimation tasks. The applicant will hire research assistants who update the estimation codes when the applicant is not in SA. We use a statistical language R for estimation, including **litedown** for reproducibility. We share the codes at the project’s Github for dissemination, transparency, and reproducibility.

Use of AI:

Limited and restrained.

- 1) As a coder: To get a hint in refining the codes to make the estimation more efficient and speedier.
- 2) As a research assistant: Conducting a literature survey.
- 3) As a logic examiner: To ask if the logic of arguments holds.

[1. Research Objectives, Research Method, etc. (continued from the previous page)]

4) As an English editor: To get suggestions on alternative ways of writing. We will then compose all sentences.

(5) Preparation status towards achievement of the purpose of the research project.

As a research program: This proposal is a sequel to the current Kakenhi funded project (23K01367) which ends this year. The applicant has been working on minimum wage research for the last 8 years. In the first paper using SARS data, we examined the producer responses to minimum wage changes. The second paper (currently ongoing research) computes HHI of firms at the local labour market level and merged them with corporate tax returns data on which we estimate minimum wage impacts. We expect a working paper to be completed this fiscal year. As a sequel, the major task of proposed research is estimating elasticities using the parametric revenue function (Raval, 2023) and the nonparametric revenue function (Pham, 2023).

Project management: The applicant, his research partner (economics professor), and his research assistants (Ph.D. and MA students in SA) have been working on the SARS data. Thanks to ongoing Kakenhi and other funding sources, we acquired not just the working knowledge but also the undocumented knowledge of data, because we sat next to the people who construct the data sets. In addition, the applicant formed a productive and professional relationship in the research team. We are registered at the Treasury Department as users of confidential data. To couple with this proposal, the applicant secured funding from the applicant's employer (2026-2028), which covers travel costs for two years and research assistant costs for three years. Jointly with this proposal, the applicant will be able to travel every year between 2026 and 2030, hire RAs for four years, and conduct research in SA.

(6) What sort of internationality this proposed research has (e.g. it will be a future world-leading research project, it will contribute to the development of global research projects through collaboration, it will create high value as a research project unique to Japan, etc.)

The proposed research is inherently international as being a bilateral project of Japan and SA. On top of this, the applicant expects further internationality in the following dimensions:

- The method is simple and widely applicable to countries with minimum wages and corporate tax returns data. This suggests a potential to spawn multilateral research comparing the effects of labour market power in disemployment impacts of minimum wages.
- The comparison of labour market power between countries can reveal new findings if linked to socioeconomic measures. Examples include: Legal framework and its implementation can reveal how a regulatory framework affects labour exploitation (wage markdowns). Juxtaposing with income inequality and unemployment can reveal how they are related to wage markdowns.

Akerberg, D. A. et al., "Identification properties of recent production function estimators," *Econometrica*, 2015, 83(6), 2411-2451.

Amodio et al., "Labor market power and self-employment around the world," 2025, working paper.

Amodio, Francesco et al., "Labor market power, self-employment, and development," *AER*, 2026 (forthcoming).

Azar, Jose et al., "Minimum wage employment effects and labor market concentration," *Review of Economic Studies*, Volume 91, Issue 4, July 2024, Pages 1843-1883

Bhaskar, Venkataraman, Alan Manning, and Ted To, "Oligopsony and monopsonistic competition in labor markets," *Journal of Economic Perspectives*, 2002, 16 (2), 155-174.

Brooks et al., "Exploitation of labor? Classical monopsony power and labor's share," *JDE*, 2021, 102627.

Callaway, Brantly et al., "Difference-in-differences with a continuous treatment," 2025, arXiv:2107.02637

Delavastita, Vincent and Michael Rubens, "Colluding against workers," *JPE*, 2025, 133.6, 1796-1839.

De Loecker, Jan, and Frederic Warzynski, "Markups and firm-level export status," *AER*, 2012: 2437-71.

Estefan, A. et al., "Outsourcing policy and worker outcomes: Causal evidence from a Mexican ban," w32024, 2024, National Bureau of Economic Research.

Faia, Ester et al., "Monopsony, markdowns, and minimum wages," NBER Working Paper 34699, 2026

Gandhi, Amit et al., "On the identification of gross output production functions," *JPE*, 2020, 128:8, 2973-3016.

Hall, Robert E., "The relation between price and marginal cost in U.S. industry," *JPE*, 1988, 96:5, 921-947.

Harasztosi, Peter and Attila Lindner, "Who pays for the minimum wage?" *AER*, 2019, 109.8: 2693-2727.

Hashemi et al., "The production approach to markup estimation often measures input distortions," *Ec Lett*, 2022.

Manning, Alan, "A generalised model of monopsony," *Economic Journal*, 2006, 116, Issue 508: 84-100

_____, "Imperfect competition in the labor market," *Handbook of Labor Econ*, 2011, IV, 973-1041.

_____, "The elusive employment effect of the minimum wage," *J. of Econ. Perspectives*, 2021, 35 (1):3-26.

Morlacco, Monica, "Market power in input markets: Theory and evidence from French manufacturing," 2019.

Pham, H., "Trade reform, oligopsony, and labor market distortion: Theory and evidence," *JIE*, 2023, 144, 103787.

Popp, Martin, "Minimum wages in concentrated labor markets," IZA Discussion Paper No. 17357, 2024.

Raval, Devesh, "A flexible cost share approach to markup estimation," *Economics Letters*, 2023, 230: 111262.

Robinson, Joan, *The economics of imperfect competition*, London, McMillan 1933.

Rubens, Michael, "Market structure, oligopsony power, and productivity," *AER*, 2023, 113 (9): 2382-2410.

Syverson, Chad, "Markups and markdowns," *Annual Review of Economics*, 2025.

Yeh, Chen et al., "Monopsony in the US labor market," *AER*, 2023, 112 (7): 2099-2138.

2. Applicant's Ability to Conduct the Research and the Research Environment

Descriptions of (1) applicant's hitherto research activities (including main research achievements), and (2) research environments including research facilities and equipment, research materials, etc. relevant to the conduct of the proposed research should be given within 2 pages to show the feasibility of the research plan by the applicant (PI) (and Co-I(s), if any).
If the applicant has conducted any international efforts related to his/her research plan (such as his/her records of joint international research and research history in overseas institutions), they should be included as necessary in "(1) applicant's hitherto research activities". Also, if the applicant has taken leave of absence from research activity for some period, he/she may choose to write about it therein.

(1) Applicant's hitherto research activities

On minimum wages and markdowns: The applicant has been studying the minimum wages and published an academic paper on the SA case in Japanese (伊藤 2021a) and several online nontechnical pieces in Japanese to inform the public (伊藤 2019a, 2019b, 2021b, 2022, 2026a). Other nontechnical piece includes 伊藤 (2026b) on wage markdowns. 伊藤 (2021a) shows statistically zero production impacts of agricultural minimum wages using various years of Commercial Agricultural Censuses. The applicant has also coauthored a paper with South African economists on agricultural minimum wages using SARS data that shows disemployment and capital intensification as the channels of firm adjustments (Kilumelume et al. 2025).

On SARS data: The applicant undertook estimation tasks at their data site: Data manipulation (e.g., reshaping to suit panel data estimation techniques), setting conditions in defining outliers, winsorization, improving efficiency of codes, parallel computing (distributing tasks across CPU cores for speeding up), and output capturing using a literate programming technique (mixing texts and code inputs/outputs in a single document) such as *quarto* and *litedown*. The applicant knows exactly what is feasible in data related tasks. The applicant also benefits from the AI use in coding tasks, albeit for static verification (without using real data) because the use of AI is not allowed on the terminals of data site.

On other micro-level data: The applicant has written and published a series of papers using micro-level data (Fuwa et al., 2006a, b, c, 2012, Ito 2006, Ito and Kono, 2009, 2010, Takahashi et al., 2017, Ito et al., 2018, 伊藤, 2021a, Ito and Shonchoy, 2026). Apart from a few papers on randomized controlled trials (RCTs), all papers rely on observational data. The applicant understands the large sample size such as SARS data is a key advantage of an observational study in providing estimates that are not suited to experimental interventions and to increase precision of estimates. The applicant has the capacity to use the most credible estimators and to efficiently write codes for computation (Ito and Shonchoy, 2026). 伊藤 (2021a) uses agriculture-climate data that the applicant matched the centroid of districts with the closest weather station using a geographical coordinate, after cleaning the scanned agricultural data with the *perl* language. The applicant firmly upholds the values of reproducible research and has the skill of automatically authoring codes in html that are published on the cloud.

Experiences in overseas collaborative research: The applicant has resided in SA and has kept close ties with the local research community for more than ten years. Acknowledging the difficulty of running an RCT on minimum wages, in 2022-2024, the applicant and the research collaborators have jointly designed a research program on minimum wages with a focus on using the administrative corporate data at SARS which culminated as a paper (Kilumelume et al., 2025). The applicant also has a record of successfully finishing studies with overseas researchers (Ito, Lepine, and Treibich 2018, Ito and Shonchoy 2026).

South African research collaborators: Prof. Dieter von Fintel, a tenured faculty member in the Faculty of Economics and Managerial Sciences, Stellenbosch University. He is a top-rated researcher who is well versed in both econometrics and SARS data manipulation. He has been working on labour issues for the last decade and has profound knowledge of local labor market functions and regulations. He has published in quality journals such as *BMJ Global Health* and *Journal of Comparative Economics* and has a piece on minimum wages using SARS data on *South African Journal of Economics* (Piek et al. 2023). He appeared on a national radio program (*The Viewpoint*, SA FM, January 23, 2019) to widely communicate about the national minimum wage impacts. There are a few more collaborators, MA and Ph.D students, a post-doc, working as research assistants who stay near the data site. They are all well versed in SARS data.

(2) Research environments including research facilities and equipment, research materials, etc.

Research environment: The applicant has exclusive access to top rated research facilities. The research collaborator, positioned in a worldly reputed university, has exclusive access to the top-rated research facilities.

Computing and coworking environment: SARS data is accessed through a lab in Pretoria. Data

[2. Applicant's Ability to Conduct the Research and the Research Environment (continued from the previous page)]

manipulation requires a fixed investment through trials and errors, and the applicant and collaborators have already invested in them. The applicant and collaborators have been working seamlessly on a common programming platform of R, while using a chat program and a video conferencing software. We have been sharing files using a file repository, GitHub. The applicant will visit SA for more intensive coding tasks.

Research materials: The applicant and research collaborators gained access to SARS data in Pretoria, South Africa.

Feasibility of research: Given past successful experiences with SARS data, it is very likely that the plan will be carried out without any roadblocks.

- Fuwa, N., Ito, S., Kubo, K., Kurosaki, T., and Sawada, Y.,** “On the Mother and Child Labor Nexus under Credit Constraints: Findings from Rural India,” *Developing Economies*, Vol.44, No.4, December 2006.
- ____ “Gender Discrimination, Intrahousehold Resource Allocation, and Importance of Spouse's Fathers: Evidence on Household Expenditure from Rural India,” *Developing Economies*, Vol.44, No.4, December 2006.
- ____ “Child Labor and School Enrollment in Rural India: Whose Education Matters?” *Developing Economies*, Vol.44, No.4, December 2006.
- ____ “How Does Credit Access Affect Children's Time Allocation? Evidence from Rural India,” *Journal of Globalization and Development*, 2012, Vol. 3, Issue 1.
- Ito, S.** “Raising educational attainment of the poor: Policies and issues,” *Developing Economies*, Vol.44, No.4, December 2006.
- Ito, S. and Kono, H.** “Publicly Provided Primary Health Care Services in Rural India: Achievements, Problems, and Possible Solutions,” in Hiroko Uchimura ed., *Health Service Provision and the Poor*, 2009, Palgrave—MacMillan, London.
- ____ “Why is Take-up of Microinsurance so Low? Evidence from a Health Insurance Scheme in India,” *Developing Economies*, Vol. 48, No.1, 2010.
- Ito, S. and Shonchoy, A. S.** “Seasonality, academic calendar and school dropouts in South Asia,” *Journal of Development Economics*, 2026, 103649.
- Kilumelume, M., von Fintel, D. Piek, M. and Ito, S.,** “Agricultural producer responses to minimum wages changes,” *Working Paper*, 2025.
- Piek, M., von Fintel, D. and Kirsten, J.,** “The impact of agricultural minimum wages on worker flows in South Africa,” *South African Journal of Economics*, 2023, 91.4: 446-465.
- Ito, S., Lepine, A., and Treibich, C.** “The effect of sex work regulation on health and wellbeing of sex workers: Evidence from Senegal,” *Health Economics*, 2018; 27: 1627-1652.
- Takahashi, K., Ito, S., Kurosaki, T., and Shonchoy, A.** “How does Contract Design Affect the Uptake of Microcredit among the Ultra-poor? Experimental evidence from the River Islands of Northern Bangladesh,” *Journal of Development Studies*, 2017, Vol. 53, No. 4: 530-547.
- 伊藤成朗, 「最低賃金引き上げの影響 (その 1)」『途上国研究の最先端 28』、IDE スクエア、2019 年 a
https://www.ide.go.jp/Japanese/IDESquare/Column/ISQ000002/ISQ000002_028.html
- ____, 「最低賃金引き上げの影響 (その 2)」『途上国研究の最先端 31』、IDE スクエア、2019 年 b
https://www.ide.go.jp/Japanese/IDESquare/Column/ISQ000002/ISQ000002_031.html
- ____, 「南アフリカにおける最低賃金規制と農業生産」『アジア経済』6 月号、2021 年 a
https://www.ide.go.jp/Japanese/Publish/Periodicals/Ajia/202106/ZAJ202106_003.html
- ____, 「最低賃金引き上げの影響 (その 3)」『途上国研究の最先端 47』、IDE スクエア、2021 年 b
https://www.ide.go.jp/Japanese/IDESquare/Column/ISQ000002/ISQ000002_047.html
- ____, 「最低賃金引き上げの影響 (その 4)」『途上国研究の最先端 62』、IDE スクエア、2022 年
https://www.ide.go.jp/Japanese/IDESquare/Column/ISQ000002/ISQ000002_062.html
- ____, 「最低賃金引き上げは低生産性労働者から高生産性労働者に雇い替えを促していた」『途上国研究の最先端 103』、IDE スクエア、2026 年 a
https://www.ide.go.jp/Japanese/IDESquare/Column/ISQ000002/ISQ000002_103.html
- ____, 「企業が共謀したのは労働者の搾取—19 世紀末ベルギー」『途上国研究の最先端 107』、IDE スクエア、2026 年 b
https://www.ide.go.jp/Japanese/IDESquare/Column/ISQ000002/ISQ000002_107.html

3. Issues Relevant to the Protection of Human Right and Compliance with Laws and Regulations

(cf. Application Procedures for Grants-in-Aid for Scientific Research)

If the proposed research involves such issues that require obtaining the consent and/or cooperation of third parties, consideration in handling of personal information, or actions related bioethics and/or biosafety, including the laws, regulations and the guidelines in the country/region(s) where the joint international research is to be conducted, describe the measures and actions planned to be taken in responding to these issues within 1 page.

This provision applies to research activities that would require approval by an internal or external ethical jury, such as research involving handling of personal information from questionnaire surveys, interviews and/or behavior surveys, including personal histories and images, handling of donated specimens, human genome analysis, recombinant DNA, and experimentation with animals. If the activities of the proposed research do not fall under such categories, enter "N/A (not applicable)".

All information used will be subject to the internal ethics code and may be subject to an approval of Research Ethics Committee (REC) of the applicant's organisation. The applicant helped establish the REC in the applicant's institution and held membership for six years, including two years served as its chair, and has substantial knowledge in research ethics and research integrity. Under the current proposal that involves anonymized data, it is unlikely that it requires an approval from the REC. Regardless of the necessity of REC approval, the applicant will make sure that all the producer information in all stages of research activities is kept anonymous, securely managed, and not shared with the third parties beyond academic purposes.

The applicant signed a nondisclosure agreement with the Treasury Department as a prerequisite to access confidential information. The applicant will comply with the laws and regulations of Japan and South Africa.

4. Items to be Entered When New Application is Made in the Fiscal Year Previous to the Final Year of the Research Period of an On-Going KAKENHI Project

(For an application that comes under this category, this column is a mandatory entry. (cf. Application Procedures for Grants-in-Aid for Scientific Research))

The applicant should give within 1 page: (1) the relevant information on the on-going project (for which FY2027 is the final year of the research period) including the original plan at the time of application/adoption and the research accomplishment such as new knowledge acquired, and (2) the reason why he/she is submitting this new proposal for FY2027 on top of the on-going project (in terms of the development of the on-going research, necessity of new research budget, etc.). If not applicable, leave this page blank. (Do not delete the page.)

Research Category	Project Number	Title of the Research Project	Research Period
			FY__ to FY2027

The original plan at the time of application/adoption and the research accomplishment of the on-going project.

The reason for submission of this new proposal.

[illegible]

設備備品費、消耗品費の必要性

ノートPC: 研究で演算および執筆に不可欠。陳腐化に備え、2年で買い換え。演算用RAMとlocal LLM用VRAMを多く搭載するために積算。

基盤研究 (C) (一般) 10-(1)
(金額単位: 千円)

[illegible]

旅費、人件費・謝金、その他の必要性

南アフリカへの現地調査：秘匿情報を使うため、作業は現地でのターミナルでしか実施できない。よって、申請者および研究助手が赴く必要がある。

その他：1. ソフトウェア(年あたり費用)、セキュリティ対策 2万円：研究倫理に照らしていかなる情報を漏洩しないため、作業を遅滞なく進めるために必須。2. AI使用料 10万円：Claude ProおよびCodexを年間契約、必要に応じてAPI追加料金支出を予定。3. サーバ設置・データ通信費用 3万円：AIを間断なく運用、並列演算機能をクラウドで実行するため。4. データ作業委託：追加的な推計に必要な計算作業を外部委託して実施するため。2万円/日、25日間。その他の年(2027・2028・2030年度)は別途予算を確保済み。